

Preventing injury

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An ounce of prevention is worth a pound of cure.

Benjamin Franklin (1706-1790)

An important role for the sports medicine clinician is to minimise activity-related injury, that is, to improve the benefit:risk ratio associated with physical activity and sport. The 2000s saw a remarkable acceleration in the focus on sports injury prevention. Fortunately, athletes can now benefit from the knowledge that interventions can prevent many major injuries, such as acute knee, ankle and hamstring injuries.

Sports injury prevention can be characterised as being 'primary', 'secondary' or 'tertiary'. In this book, we use the term 'prevention' synonymously with what is technically known as 'primary prevention'.¹ Examples of primary prevention include health promotion and injury prevention (e.g. ankle braces being worn by an entire team, even those without previous ankle sprain). Secondary prevention can be defined as early diagnosis and intervention to limit the development of disability or reduce the risk of re-injury. We refer to this as 'treatment' in this book (e.g. early treatment of an ankle sprain; see Chapter 17). Finally, tertiary prevention is the focus on rehabilitation to reduce and/or correct an existing disability attributed to an underlying disease. We refer to this as 'rehabilitation' (Chapter 18); in the case of a patient who has had an ankle sprain, this would refer to balance exercises and graduated return to sport after the initial treatment for the sprain.

The proactive clinician will give prevention advice during consultations where treatment is being sought. Health professionals working with teams will initiate injury prevention strategies, whether engaged in a traditional team sport like football, or working with a team of individual athletes such as in athletics or swimming. To guide this process, this chapter will introduce the concept

of risk management,² which includes systematic injury surveillance, pre- and in-season strategy planning sessions with coaches, regular screening of athletes (see also Chapter 46), and the development of specific prevention programs for the team and for the individual athlete.

From there, this chapter will cover specific topics in injury prevention, including:

- prevention of hamstring strains, ankle sprains and acute knee injuries
- prevention of overuse injuries
- managing load to prevent injury
- protective equipment
- appropriate surfaces.

A CONCEPTUAL APPROACH TO INJURY PREVENTION

Willem van Mechelen et al. introduced the now classic conceptual model for research on sports injury prevention (Fig. 12.1).³ This model can be successfully applied by sports medicine clinicians as well. First, identify the magnitude of the problem and describe it in terms of the incidence and severity of sports injuries. If you are responsible for a team, this involves recording all injuries within the squad, as well as training and match exposure. Second, identify the risk factors and injury mechanisms that play a part in causing these sports injuries. For the clinician, this involves systematic steps to examine the athletes, and their training and competition program (see below). The third step is to introduce measures that are likely to reduce the future risk and/or severity of sports injuries, based on the aetiologic factors and injury mechanisms identified in the second step. Finally, the

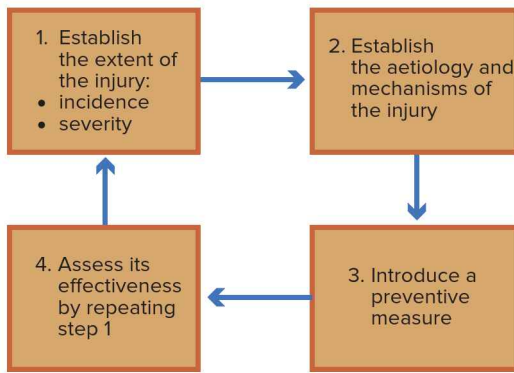


Figure 12.1 A conceptual model of injury causation³

FROM VAN MECHELEN ET AL

effect of the measures must be evaluated by repeating the first step. In the research setting, preventive programs are best evaluated using a randomised controlled trial design. For the clinician responsible for a team, continuous surveillance of the injury pattern within the team will reveal whether changes occur in the injury rate.

Clinicians wishing to prevent injuries in a systematic way could base their approach on the updated model by Meeuwisse et al. to describe the potential causative factors for injury.^{4, 5} This conceptual model was also expanded by Bahr and Holme⁶ and Bahr and Krosshaug⁷ (Fig. 12.2). The model illustrates the multifactorial nature of sports injuries, emphasising the relationship of intrinsic and extrinsic risk factors to injury mechanisms.

First, it considers the intrinsic risk factors—factors which may predispose or protect the athlete from injury. This includes athlete characteristics such as age, maturation, gender, body composition and fitness level. One factor that consistently has been documented to be a significant predictor almost regardless of the injury type studied is previous injury. Intrinsic factors such as these interact to predispose to or protect from injury.

Intrinsic risk factors can be modifiable and non-modifiable, and both are important from a prevention point of view. Modifiable risk factors may be targeted for change (e.g. through specific training methods). Non-modifiable factors (such as sex) can be used to target intervention measures to those athletes who are at increased risk. An example of the clinical relevance of sex as a non-modifiable risk factor in sports injury prevention is the higher risk of injury to the anterior cruciate ligament (ACL) in female athletes compared with males. Female athletes are at up to six times greater risk for sustaining an ACL tear than their male counterparts in comparable team sports.⁸

The second group of risk factors is the extrinsic factors athletes are exposed to; for example, floor friction in indoor team sports, snow conditions in alpine skiing, a slippery

surface (running track), very cold weather, or footwear. In the same way as intrinsic factors, extrinsic factors may increase risk (e.g. inappropriate footwear) or protect from injury (appropriate footwear). Exposure to such extrinsic risk factors may interact with the intrinsic factors to make the athlete more or less susceptible to injury.

When intrinsic and extrinsic risk factors act simultaneously, the athlete may be at far greater risk of injury than when risk factors are present in isolation. Note that risk factors—intrinsic (e.g. muscle strength) and extrinsic (e.g. weather conditions)—are usually not static, but change with time, and can even change during one training session (e.g. due to fatigue).

The inciting event

The final component in injury causation is the inciting event, which is usually referred to as the injury mechanism—what we see when watching an injury situation. Again, it may be helpful to use a comprehensive model to describe the inciting event, which accounts for the events leading to the injury situation (playing situation, player and opponent behaviour), as well as to include a description of whole body and joint biomechanics at the time of injury (Chapter 8).⁷ Each injury type and each sport have their typical patterns, and for team medical staff it is important to consult the literature to reveal the typical injuries and their mechanisms for the sport in question. However, one limitation of the model is that it is not obvious how the team's training routine and competitive schedule can be taken into consideration as potential causes, and the model has therefore traditionally mainly been used to describe the causes of acute injuries. For overuse injuries, the inciting event is usually distant from the outcome. For example, for a stress fracture in a long-distance runner, the inciting event is not usually the single training session when pain became evident, but the training and competition program he or she has followed over the previous weeks or months.

RISK MANAGEMENT: APPLYING PREVENTION MODELS TO YOUR TEAM

For clinicians, the conceptual models can be adapted to identify potential causes of injury and develop a targeted prevention program. The key questions to ask are: What are the typical injuries? Who is at increased risk? Why? And how do injuries typically occur?

When caring for a defined group of athletes such as a football team or an alpine skiing team, this can be done using a systematic risk management approach.² This includes:

- reviewing the literature—what are the typical injuries patterns in the sport?

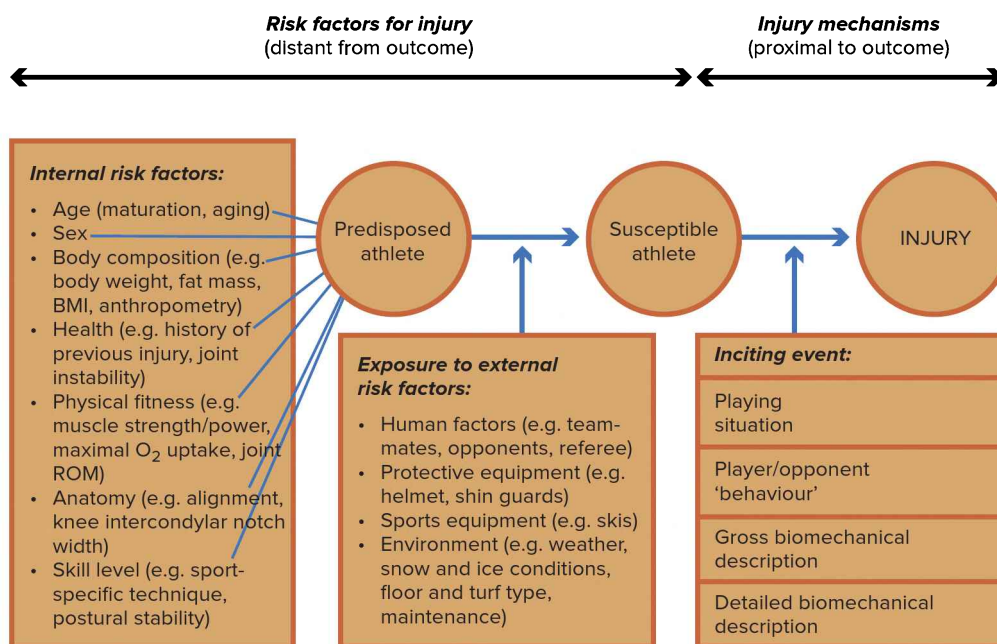


Figure 12.2 A comprehensive injury causation model based on the epidemiological model of Meeuwisse et al.⁴ and modified by Bahr & Krosshaug.⁷ BMI = body mass index; ROM = range of motion

- developing an injury surveillance program within the team—recording injury and participation data
- season analysis—risk profiling the training and competition program
- performing a periodic medical assessment—mapping current problems and intrinsic risk factors
- developing and initiating a targeted prevention program.

Reviewing the literature—risk identification and assessment

Each sport has its typical injury pattern. Just think of the names of some sports injuries: tennis elbow, runner's knee, jumper's knee. For most sports, there is ample data in the literature to identify and assess the risks. Note that injury risk is not just a question of injury frequency—the severity of injury must also be taken into account.

Injury data can be illustrated by a risk matrix that highlights risks in terms of likelihood and consequences. A risk matrix is a powerful tool for risk assessment. We derive the example shown in Figure 12.3 from soccer. It suggests that injury reductions in the areas of ACL, hamstring and ankle injuries are priorities. By further examining which factors contribute to the causation of these, we can formulate strategies to reduce injury rates. Therefore, in soccer it makes sense to target training programs to prevent ACL tears towards female athletes,

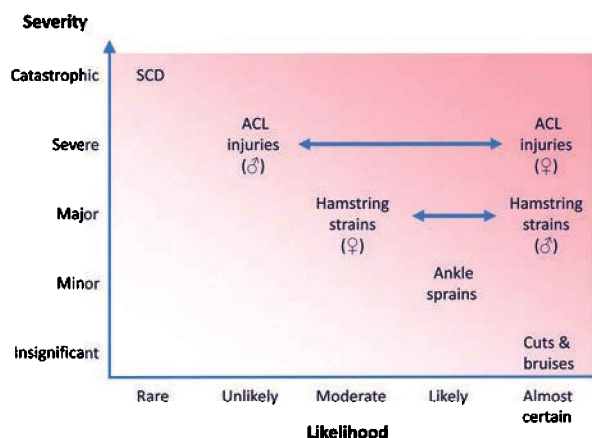


Figure 12.3 Qualitative risk matrix in elite soccer, illustrating the relationship between injury severity (consequence) and injury incidence (likelihood). The darker the colour, the greater the cross-product of severity and incidence, and the greater the priority should be given to prevention. The matrix also illustrates that risk differs between males and females. (SCD = sudden cardiac death)

while the lower risk among males may not justify such interventions. In contrast, a prevention program for male players should definitely focus on preventing hamstring strains.

Developing an injury surveillance program within the team

Although basing the risk assessment on data from the literature is valuable, it is necessary to establish a system to monitor injuries and exposure to assess risk within your own team. For medical staff, recording injuries is one of the easiest tasks in risk management as we are required to keep accurate records of all assessments and treatment provided to our patients. Thus, establishing a surveillance system simply involves analysing information that is already there. To make this task even easier, excellent electronic tools are available, with the necessary statistical tools integrated in software used to keep routine patient records.

Another task is to establish a system to record individual training and competition exposure within the team. This is a challenge for the medical team, since they are not always present during practices or on road trips. Therefore this task is often done by the coaching/fitness staff. Many of the injury-recording software programs also have the capability of recording exposure data, which can then be entered based on the coaching records. Exposure data is necessary to be able to calculate risk.

The standard method for calculating injury incidence typically used in team sports is the number of time-loss injuries per 1000 hours of exposure. For some sports and specific player positions (e.g. a baseball pitcher), injuries per pitch may be a more appropriate measure when identifying relationships between injury and exposure. It may also be relevant to record exposure in relation to extrinsic risk factors, such as turf type (e.g. training on grass, gravel, artificial turf), use of personal protective

equipment, and whether the exposure is during competition or training.

Interpretation of injury data is difficult unless exposure data are collected. Consider the example where there is an increase of 30% in the number of match injuries from one season to the next. If the number of matches also increased by 30%, then there would be no increase in injury incidence.

A limitation with the standard methods for injury registration is that they substantially underestimate the true burden of overuse injuries due to a reliance on time-loss injury definitions. Overuse injuries represent the predominant injury type in sports that involve long, monotonous training sessions (e.g. cycling, swimming and long-distance running), as well as in technical sports that involve the repetition of similar movement patterns such as throwing or jumping. Symptoms such as pain or functional limitation most often appear gradually and may be transient in nature. Therefore it is likely that athletes will continue to train and compete despite the presence of overuse conditions, at least in the early phase. The same is the case for minor illness, such as the common cold.

Thus many health problems, especially overuse problems, do not lead to time loss from sport and are therefore not recorded in standard injury surveillance systems.⁹ To address this limitation, Clarsen et al.^{10, 11} have developed the Oslo Sports Trauma Research Center (OSTRC) Questionnaire on Health Problems for prospective monitoring of all illness and injury, not just those leading to time loss from sports participation. This approach may be more appropriate for sports where overuse injuries and illnesses are expected to dominate.

Oslo Sports Trauma Research Center Questionnaire on Health Problems

The OSTRC Questionnaire on Health Problems is distributed to athletes once a week to record and monitor the consequences of any illnesses and injuries they have incurred, including those that do not lead to time loss from sport.¹⁰ Ideally, it is delivered as an electronic questionnaire that directs athletes only to relevant questions, based on their responses.

The questionnaire begins with **four key questions** that all athletes should answer (Table 12.1). If they indicate a health problem in any of the questions (i.e. any response except the minimum value), they are then asked:

- whether they are referring to an illness or an injury
- to provide the location (for injuries) or the symptoms (for illnesses)
- to indicate the number of days of time loss the problem has caused in the past week
- whether they have received medical attention for the problem
- whether they have any more problems (in which case the questionnaire starts again).

In some teams and organisations, this weekly questionnaire can act as an important communication platform between athletes and sports medicine clinicians, particularly when athletes are geographically dispersed and/or have inconsistent contact with their team's medical staff. In these cases, it may be useful to add extra questions to the questionnaire such as 'who knows about this problem?' and 'do you have any comments for your team medical staff?'

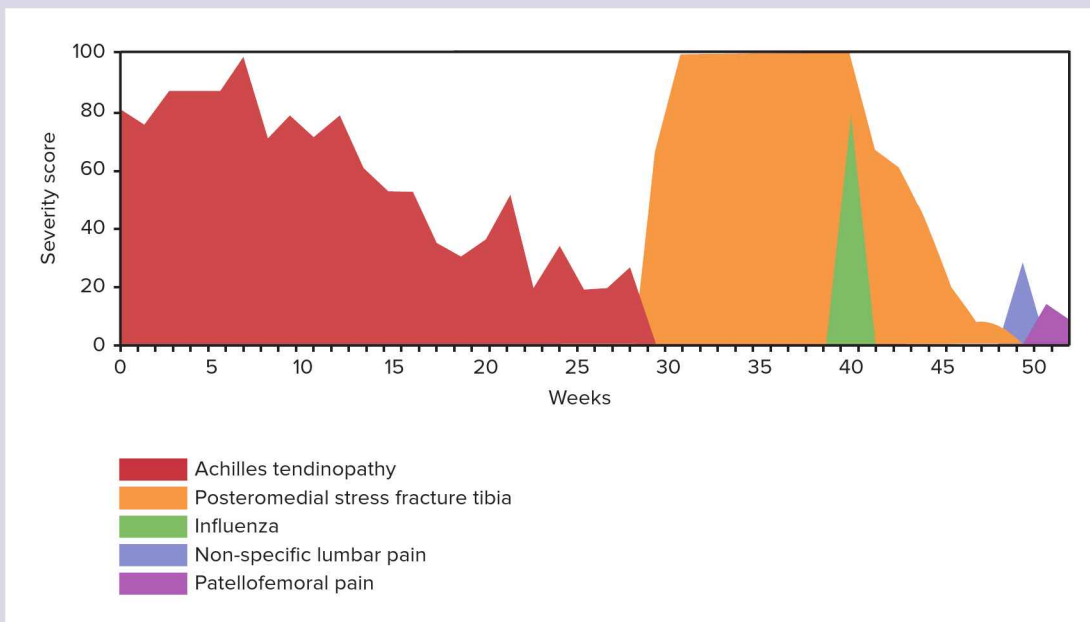
The severity of each reported problem is rated using the scoring system shown in Table 12.1. By adding the scores for Questions 1 to 4, a severity score out of 100 can be calculated. This score is tracked over time to give a reflection

of the overall impact of the problem and of fluctuations in its severity (Fig. 12.4a). Summary data for the entire team can be collected (Fig. 12.4b).

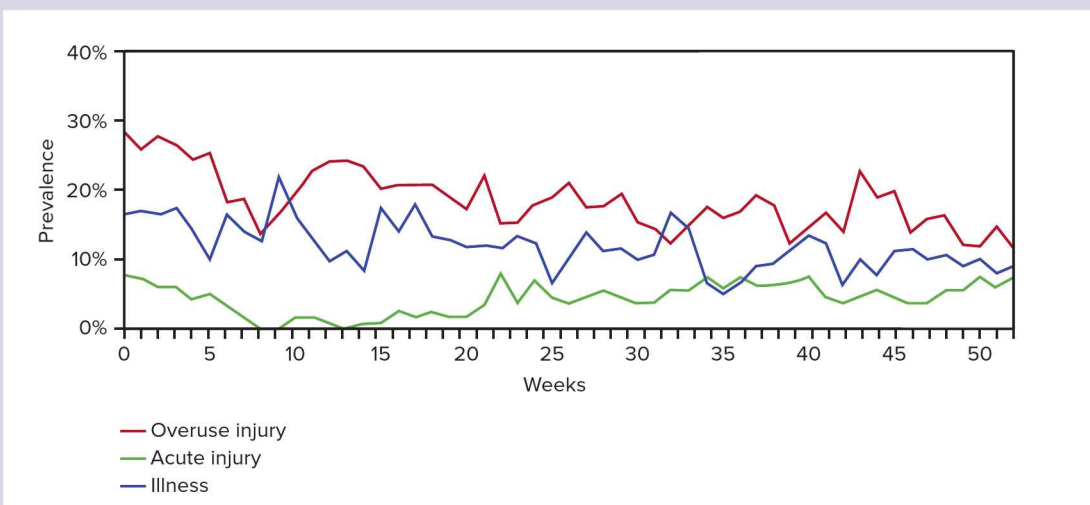
Table 12.1 Oslo Sports Trauma Research Center Questionnaire on Health Problems

Participation	Severity score
Q1. Have you had any difficulties participating in normal training or competition due to injury, illness or other health problems during the past week?	
• Full participation without health problems	0
• Full participation, but with injury/illness	8
• Reduced participation due to injury/illness	17
• Cannot participate due to injury/illness	25
Training volume	
Q2. To what extent have you reduced your training volume due to injury, illness or other health problems during the past week?	
• No reduction	0
• To a minor extent	6
• To a moderate extent	13
• To a major extent	19
• Cannot participate at all	25
Training volume	
Q3. To what extent has injury, illness or other health problems affected your sports performance during the past week?	
• No effect	0
• To a minor extent	6
• To a moderate extent	13
• To a major extent	19
• Cannot participate at all	25
Training volume	
Q4. To what extent have you experienced symptoms/health complaints (e.g. pain, coughing, fever) during the past week?	
• No symptoms/health complaints	0
• To a minor extent	8
• To a moderate extent	17
• To a severe extent	25

continued



(a)



(b)

Figure 12.4 Example data from injury surveillance using the Oslo Sports Trauma Research Center (OSTRC) Questionnaire on Health Problems (a) Individual athlete data (b) Team data showing prevalence

Season analysis—risk profiling the training and competition program

One helpful method to manage risk in sports is a pre-season review of the planned training and competition program to identify risks. The method of season analysis therefore is fundamentally different from injury

surveillance, where data on injuries are collected as they happen. Season analysis represents an attempt to identify risks before they occur. Risks in the program can be related to the competition schedule, the training program, the possibilities for athlete recovery, travel or other issues.

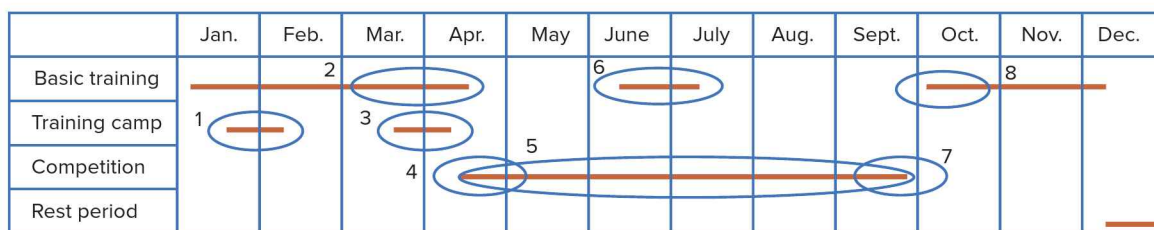


Figure 12.5 Risk profile. Examples of periods of season when a college basketball team may be at particular risk of injury. The comments below concern the risk periods that are circled.

1. Change of time zone, off-court training surface, climate and altitude during training camp in Colorado. Emphasis on defensive stance training and quick lateral movements could lead to several groin injuries (Chapters 31 and 32). Athletes should not increase the amount or intensity of training too much.
2. Transition to greater amount of on-court training and intensity, combined with several practice games. Floor surface quite hard. Risk of lower limb injuries such as Achilles tendinopathy, medial tibial stress syndrome.
3. New training camp to fine-tune players before beginning the competitive season; practice games on unusually slippery courts. Competition to avoid being cut from the squad leads to increased intensity during training and competition.
4. The beginning of the competitive season. A higher tempo and a packed competitive schedule to which the athlete is unaccustomed. Risk of overuse injury (e.g. patellar tendinopathy, tibial stress fracture) compounded by heavy academic program leads to additional fatigue.
5. High risk of acute injuries during the competitive season, and a tough competition schedule at full intensity.
6. Interposed period of hard basic training, with strength exercises to which the athlete is not accustomed, and plyometric training increases risk of tendinopathy and muscle strain.
7. The end of the competitive season. Worn out and tired players? This is an important time to treat low-level 'grumbling' injuries aggressively. Waiting for the injury to heal with 'rest' alone is not recommended.
8. Transition to basic training period with running on trails.

The analysis is based on the idea that the risk of injuries is greater during transitional periods and that each stage has certain characteristics that may increase risk. Examples are when athletes switch from one training surface to another (e.g. from grass to gravel) or to new types of training (e.g. at the start of a strength training period). Figure 12.5 illustrates how a team may be at particular risk for different injury types at different stages of the season. Other examples of key events which could be correlated with increased injury risk include:

- poor sleep due to tight schedule or time differences
- changeover from heavy pre-season training to competition
- return to play after mid-season pause
- beginning of final rounds
- increased training and competition load associated with representative duties
- change of coach/manager with different training methods
- change in training volume
- change of climate (e.g. move from a training camp in a warm climate to a colder climate)
- selection time for important matches (e.g. representative schedule—a player may hide early symptoms of an injury, thinking this may prevent selection).

Although medical personnel responsible for teams or training groups may have to initiate this type of analysis,

it is strongly recommended that the process be done in collaboration with the coaching team and, if at all possible, the athletes. Their inclusion will enable them to draw on their past experiences with the team, which is especially important if there are no injury surveillance data available from the past. If injury surveillance data are available, the season analysis is an opportunity to review past experiences formally, and discuss whether the injury patterns seen may be related to the training and competition program. For example, a surge in stress fractures in a soccer team may be attributed to a simultaneous increase in the volume of running and a change from a soft to a hard running surface. This type of analysis is an important basis for planning preventive measures, particularly to avoid overuse injuries. The risk profile usually varies from sport to sport, which underlines how important it is that medical staff be intimately familiar with the characteristics of the sport they cover.

The periodic medical assessment—mapping current problems and intrinsic risk factors

Periodic medical assessments (PMAs) are routinely performed on hundreds of thousands of athletes around the world every year, in some cases required by sports regulations or even by law.

PRACTICE PEARL

The value of doing routine physical exams on athletes is questionable. If done properly, they can represent a key ingredient in the risk management program. If they are done simply to clear athletes for participation, their value in injury prevention is limited. Unfortunately, this is commonly the case.

Chapter 46 provides a detailed discussion of the many potential objectives of performing a PMA. In a prevention context, two of these are particularly relevant:

1. to identify current illnesses, injuries and chronic medical conditions
2. to identify factors that increase risk of future injury or illness.

While a PMA is often referred to as a pre-participation or pre-season examination, it is better performed at the end of the season, when there is still sufficient time for optimal treatment and rehabilitation of ongoing illness or injury and time to work on correcting any risk factors identified.

The factors included in the PMA should be tailored to the sport in question, by focusing on conditions known to be particularly prevalent (e.g. low back problems in rowers, eating disorders among female gymnasts, reduced pulmonary function among cross-country skiers), or where the physical requirements of the sport imply a higher risk of certain conditions (e.g. Marfan syndrome among basketball or volleyball players). However, if the purpose is to identify risk factors to predict future injury, the value of doing routine examinations alone is limited. For screening exams to serve a purpose in risk management, there are some additional requirements. First, the exam must be designed to identify athletes with risk factors relevant to the sport in question. Second, there must be a plan to follow up athletes with measures intended to reduce risk, if risk factors are identified. Third, the screening exam and follow-up must be planned and led by the medical and coaching staff of the team. Let us consider each of these requirements in turn.

1. Identify athletes with risk factors

The first requirement, that the exam be designed to identify athletes with risk factors relevant to the sport in question, is rarely met in current practice. In fact, in most cases the same screening exam is used across sports. This is inappropriate. There can be no single recipe for all sports, as injury patterns and risk factors differ significantly. To design a screening program for one particular sport, it is therefore necessary to define the key injury types (based on data from the literature on incidence and severity, and preferably also surveillance data as described earlier),

and then to define the key risk factors (for these injuries). Based on this information, the screening exam can be tailored to the sport.

Even if the key risk factors have been defined, a key task remains—to select appropriate methods to screen for these risk factors. Screening methods should be valid, accurate, sensitive and specific. Valid means that the testing method measures the factor you wish to measure. For example, if low hamstring strength is a risk factor for muscle strains, what method should be used to measure this best? That the test is accurate (reliable), means that it will yield the same result each time. Finding valid and accurate tests to measure relevant risk factors is difficult.

However, to predict who may be at increased risk for injury is nearly impossible. A testing method with high sensitivity will identify all players with increased risk. A test with high specificity will identify only players with increased risk. Few, if any, tests exist which can clearly distinguish between athletes with high and low risk for a sports injury.¹² In fact, no tests exist which have high sensitivity *and* high specificity to predict future injury. Therefore, screening tests should be interpreted with caution.

2. Plan to follow up risk factors

The second requirement is that if the presence of a known injury risk factor is identified, an appropriate course of action can be taken to control the risk. If strength is inadequate, it should be followed up with a strength training program. If balance is poor, a training program should be designed and implemented to rectify this. This might be on an individual level, but in the absence of predictive screening tests, a team approach is probably more appropriate.

3. Medical and coaching staff of the team must be involved in screening and follow-up

In community-level sport, it may be tempting for a club or school to enlist the assistance of local sports medicine personnel to screen their athletes. This is only advisable if they can establish a mechanism to communicate the results to the team, and there are resources to follow up athletes with health problems or risk factors. Also, the specialist must be up to date with screening methods for the sport in question and their interpretation.

Developing and initiating a targeted prevention program

The final step in the risk management process is to assess the risks to the team and the individual athlete. The injuries (and illnesses) to be targeted are identified from the risk matrix (Fig. 12.3), the scientific literature and team injury surveillance data from past seasons. The season analysis may have identified specific risks associated with the training and

competition program. Finally, the specific injury and risk profile for the individual athlete is mapped through the PMA. Based on this assessment, a prevention program targeting these risks should be developed for the team and for the individual athlete. The next sections provide examples of prevention programs targeting three acute sports injuries—hamstring strains, ankle sprains and acute knee injuries.

PREVENTING HAMSTRING STRAINS

Hamstring strains are frequent in sports involving maximal sprints and acceleration. Among sprinters, hamstring strains represent approximately one-third of all acute injuries.¹³ In studies from professional sports, hamstring strains rank as the first or second most common injury in soccer,^{14–17} Australian rules football,^{18, 19} rugby union and league,^{20, 21} and American football,^{22, 23} accounting for one in every five to six injuries in most studies. There also seems to be a trend with a gradual increase in the proportion of hamstring strains compared to other injury types.^{24, 25} Hamstring strain injuries are also common in dancing and waterskiing.²⁶

Injury mechanisms

Hamstring strains most often occur during maximal sprints. It is difficult to document exactly at what time during the running cycle injuries occur.²⁷ However, since the net moment developed by the hamstrings is thought to be maximal in the late swing phase right before heel strike, this is thought to be a vulnerable position.^{28, 29} In this instance, the hamstring muscles work eccentrically. Another suggestion is at push-off.

Risk factors

A number of risk factors have been proposed for hamstring strains, the most prominent being the following four intrinsic factors: age; previous injury; reduced hip range of motion (ROM); and poor hamstring strength.^{6, 30} In theory, limited ROM for hip flexion could mean that muscle tension is at its maximum when the muscle is vulnerable close to maximum length. However, this hypothesis has yet to be confirmed, since there are several studies on soccer players suggesting that hamstring flexibility is not a risk factor for strains.^{31, 32}

Low hamstring strength would mean that the forces necessary to resist knee flexion and initiate hip extension during maximal sprints could surpass the tolerance of the muscle-tendon unit. Hamstring strength is often expressed relative to quadriceps strength as the hamstrings:quadriceps ratio, since it is the relationship between the ability of the quadriceps to generate speed and the capacity of the hamstrings to resist the resulting forces that is believed to be critical. Several studies show that players with low hamstring strength or low hamstrings:quadriceps strength ratio or side-to-side strength imbalances may be at increased risk of injury.⁶

A history of previous hamstring strain greatly increases injury risk, as documented in numerous studies.^{6, 33, 34} Injury can cause scar tissue to form in the musculature, resulting in a less compliant area with increased risk of injury. A previous injury can also lead to reduced ROM or reduced strength, thereby indirectly affecting injury risk. Soccer players with a history of previous hamstring injury have a 5–7 times higher risk of injury.

Older players are at increased risk for hamstring strains, and although older players will be more likely to have a previous injury, increased age is also an independent risk factor for injury.^{32, 34}

Short fascicle length in the long head of the biceps femoris is an independent risk factor, especially when combined with low levels of eccentric knee flexor strength.³⁵ Notably, the greater risk of a future strain injury in older players or those with a previous hamstring injury is reduced when they have longer fascicles and high levels of eccentric strength.

It has been suggested that an indicator of susceptibility for the damage from eccentric exercise is the optimum angle for torque.³⁶ When this is at a short muscle length, the muscle is thought to be more prone to eccentric damage, but this theory has been challenged.³⁷

Other risk factors, which have been suggested but are less well studied include race, sex, level of play, player position, improper running technique, superior running speed (peak performance), low back pain, increases or changes in the training program (particularly intense periods of training), insufficient warm-up and muscle fatigue. Players of African or Australian Aboriginal origin sustain significantly more hamstring strains than Caucasian players,³⁴ and it has been suggested that these players may be faster runners when compared to their Caucasian counterparts, possibly because of a higher proportion of type II muscle fibres. A faster running speed will generate higher hamstring torques, which may explain the increased injury risk.

Prevention programs

There are no intervention studies on elite athletes on the preventive effect of flexibility training on hamstring strains. However, one study on military basic trainees indicates a reduced number of lower limb overuse injuries after a period of hamstring stretching,³⁸ while another military-based study found no effect of stretching.³⁹ However, it should be noted that these studies were designed to examine the effect of general stretching on lower limb injuries in general, not a specific hamstring program on hamstring strain risk.

Questionnaire-based data on flexibility training methods collected from 30 English professional football clubs, where the stretching practices of the teams were correlated to their hamstring strain rates, indicate that using a standard stretching protocol reduces injury risk.⁴⁰ A study from Australian rules football has observed a

reduction in the incidence of hamstring strains with a three-component prevention program, where stretching while fatigued was one of the components.⁴¹ The other factors in the program were sport-specific training drills and high-intensity anaerobic interval training. Thus, it is not possible to determine which of these factors are responsible for the observed effect.

The best evidence for injury prevention is available for programs designed to increase hamstring strength, particularly eccentric hamstring strength. Some studies indicate that low hamstring strength may be a risk factor for sustaining hamstring strains,^{31, 42, 43} particularly eccentric strength.^{44, 45} EMG studies have shown that activity is highest late in the swing phase and during heel strike, when the hamstrings work eccentrically or transfer from eccentric to concentric muscle action.^{29, 46} It is assumed that most hamstring strains occur during eccentric muscle actions, when the muscle activity is highest.^{34, 35} It is well documented that strength training is mode specific.^{47, 48} Based on this it may be argued that, to be specific, strength training for the hamstring muscles should be eccentric.

Recent studies from Scandinavia have shown that replacing the traditional hamstrings strength exercise used by teams—hamstring curls—with exercises to develop eccentric strength reduces the risk of hamstring strains.^{14, 49} Traditional hamstring curls were ineffective in increasing eccentric hamstring strength among elite athletes.⁵⁰ In contrast, a simple partner exercise—the Nordic hamstring lower (Fig. 12.6)—improved eccentric strength.⁵⁰ A pilot study has also shown that using a special apparatus—the Yo-Yo™ flywheel ergometer—also increases eccentric hamstring strength.⁵¹ Both of these methods prevented hamstring strains in studies on soccer players^{14, 49–52} and rugby union players.²⁰

Because it is easy to implement Nordic hamstring exercises in a team setting, we recommend this exercise as a specific tool to prevent hamstring injuries. However, to avoid delayed onset muscle soreness it is important to follow the recommended exercise prescription with



Figure 12.6 The Nordic hamstrings exercise. Subjects are instructed to let themselves fall forwards, and then resist the fall against the ground as long as possible by using their hamstrings

a gradual increase in training load when introducing a program of Nordic hamstring lowers (Table 12.2). By the end of a 10-week training period, many players are able to stop the downward motion completely before touching the ground (i.e. at about 30° of knee flexion), even after being pushed by their partners at a considerable speed. When a player can reach this stage, the characteristics of the Nordic hamstring lower exercise appear to resemble the typical injury situation: eccentric muscle action, high forces and near full-knee extension. The program has been implemented in several different sports and younger age groups, and injuries from the exercise itself have not been recorded.

PREVENTING ANKLE SPRAINS

with EVERT VERHAGEN

Ankle sprains are the most common injuries seen in sports and recreational activities. Nearly every fifth patient seen in the emergency room after sports-related injuries has

Table 12.2 Training protocol for Nordic hamstring exercises (see Fig. 12.6).⁵⁰ Load is increased as subject can withstand the forward fall longer. When managing to withstand the whole ROM for 12 reps, increase load by adding speed to the starting phase of the motion. The partner can also increase loading further by pushing at the back of shoulders

Week	Sessions per week	Sets and repetitions
1	1	2 × 5
2	2	2 × 6
3	3	3 × 6–8
4	3	3 × 8–10
5–10	3	3 × 12/10/8 reps

sustained an ankle sprain.⁵³ It has been estimated that there is one ankle sprain per 10 000 persons each day, which translates to approximately 10 million sprains every year in the United States alone.⁵⁴ It has been estimated that about 25% of all injuries across all sports are ankle injuries, and even more in some sports like volleyball, basketball and orienteering.⁵⁵⁻⁶⁰

Injury mechanisms

The typical injury mechanism is landing or stepping with the foot in an inverted position (i.e. plantarflexed, internally rotated and supinated). With the foot in this position, the ankle joint is inherently unstable. The posterior talar plafond is more narrow than the anterior portion, thereby reducing the bony stability of the ankle mortise when the foot is plantar flexed. Unless the dynamic musculotendinous ankle stabilisers can compensate for this reduced structural stability when the ankle joint is perturbed, the ligaments that statically stabilise the lateral ankle are acutely overloaded when the athlete puts weight on the inverted foot. In soccer, such perturbations often result from tackles, where the player receives a laterally directed hit on the medial side of the ankle or lower leg, whereupon landing in a supinated position leads to an inversion sprain.^{55, 56} In volleyball and basketball, injuries often result from landing on the foot of an opponent or teammate.⁵⁷⁻⁵⁹

Risk factors

The most important risk factor that has been identified for ankle sprains is a previous ankle injury in the past 12-24 months. In fact, research has shown that among adult athletes, four out of five ankle sprains occur in previously injured ankles.^{59, 61, 62} Compared to an ankle with no prior injury, the risk of injury is fourfold greater for an ankle that has been sprained one or more times.⁵⁷ Furthermore, the more recent the injury, the higher the risk. The injury rate during the first 6-12 months after an ankle sprain is nearly tenfold higher than for an ankle without previous injury. Following an ankle sprain, the proprioceptive system of the ankle is affected, hampering the stability and sense of movement of the ankle joint. This subsequently leads to more inverted positions upon landing from a jump or stride, as well as lesser control over the dynamic musculotendinous ankle stabilisers to counteract perturbations, leading up to a higher chance of subsequent injury in equal situations.^{63, 64}

Postural sway is closely linked to a previous ankle sprain occurrence.⁶⁵ Postural sway is generally characterised in a practical approach that involves measurements of the duration of time that a subject can maintain a single-leg stance without touching down to recover balance. Athletes who are able to maintain a single-leg stance for

at least 15 seconds are considered to have normal posture, while those that touchdown to regain balance within the 15-second test are considered to have abnormal posture. Athletes with abnormal posture are more likely to suffer ankle sprains.

A number of other risk factors have been suggested to be associated with an increased injury risk, such as sex, height and weight, limb dominance, foot type, foot size, anatomic alignment of the lower extremity, ankle ROM, ankle and generalised joint laxity, and muscle strength. However, the relationship between these factors and injury risk appears to be weak.⁶⁵

Prevention programs

There is strong evidence that taping or bracing should be recommended for a period of up to 12 months after an ankle sprain, when risk of injury is increased several times.⁶⁶ The mechanism by which such ankle orthoses are thought to work is not known with certainty, but may involve simply enhancing the athlete's proprioceptive awareness of the ankle joint. This view is corroborated by the fact that the preventive effect of braces is limited to players with previous injury, where proprioceptive function is reduced.⁶⁷⁻⁷⁰ In addition, orthoses do not seem to restrict ankle inversion enough to explain their prophylactic effect on ankle sprain incidence. If the protective effect were purely mechanical, one would expect an effect in healthy, previously uninjured ankles as well. The latter has been shown in basketball, but this finding remains to be confirmed in additional research.⁷¹ Many different ankle supports are commercially available.

Ankle taping restricts inversion motion, although it appears that ankle supports are superior to ankle taping since ankle supports do not lose their ability to restrict inversion, while tape does 'loosen up' after several repeated cycles of vertical jumping. Unlike semi-rigid orthoses, the effectiveness of ankle taping has not been tested in randomised controlled trials, but if the effect is mainly through enhancement of proprioception, there is no reason to expect taping has less preventive effect than orthoses. Other factors, such as cost and skin care, should also be considered in the choice between tape and orthoses. Finally, there is no evidence that wearing an ankle orthosis increases the incidence of knee injuries, and most studies suggest that semi-rigid orthoses do not significantly impair athletic performance.⁷²

Tropp et al.⁶⁸ and others^{69, 70} have shown that proprioceptive function is reduced in athletes who complain of a feeling of persistent instability following an ankle sprain. Sensorimotor control of the affected ankle joint is impaired in the immediate recovery period following an acute sprain,^{63, 64} but studies have shown that this function can



Figure 12.7 Balance board training

be restored through a balance board training program.^{73–75} In these studies, proprioceptive function was quantified by measuring the reaction time to a sudden inversion strain, or the degree of postural sway during a one-legged balance test. It should be noted that the use of the term proprioceptive function, which is defined as the function of the afferent components only, may be inappropriate in this context. The ability to react to a sudden inversion stimulus or balance on one leg clearly depends on both sensory and motor function, and should perhaps therefore be termed sensorimotor control.

The preventive effect of balance and sensorimotor exercise has been well described in the literature.^{66, 76} A balance board training program has been shown to reduce the risk of re-injury in functionally unstable ankles in soccer players.⁶⁸ This basic program consists of balance exercises on one leg on a disc (Fig. 12.7). Over the years, this basic program has evolved into a variety of sports-specific multifaceted sensorimotor training programs that involve more types of exercises than one-legged balance exercises alone. In a one-season randomised trial, for example, balance exercises were introduced as part of the regular warm-up

for high-level recreational volleyball players.⁷⁷ Ankle sprain incidence was reduced by 60% among players who reported a history of previous injury. Other effective examples can be found for other sports including soccer,⁷⁸ handball,⁷⁹ basketball⁸⁰ and American football.⁸¹

The specific preventive effect of sensorimotor training has also been studied as part of the rehabilitation process after an initial ankle sprain. This yielded positive results not only for the prevention of secondary injury,⁸² but also for other complaints that may persist or develop after the initial ankle injury, such as pain and instability.⁷⁴ Consequently, clinical care guidelines advocate the application of sensorimotor training within or straight after the rehabilitation phase after an ankle sprain.⁸³

The exact intensity and duration of the program depends on the individual patient and the severity of the initial ankle sprain.

PRACTICE PEARL

It appears reasonable to recommend a program of 10–20 minutes of balance board training five times a week over 10–12 weeks for all athletes with a history of ankle sprain (Fig. 12.7).

In other words, taping, bracing and neuromuscular training are all effective for the prevention of ankle sprain recurrences. Although preventive effects have been reported in a general athletic population, evidence suggests this overall effect is due to a strong preventive effect in previously injured athletes (tertiary prevention) and that any effect on fresh ankle sprains (primary prevention) is either non-existent or very low.

The next question is then: which measure should be preferred for the prevention of ankle sprain recurrences? Naturally, the answer is that the method the athlete prefers will achieve the best results. After all, this is the measure that the athlete is most likely to use. However, a combination of an extrinsic prophylactic measure (tape or brace) with neuromuscular training will probably achieve the best preventive outcomes with minimal burden for the athlete (Fig. 12.8).

The preventive potential of combining bracing/taping with balance training lies in the different pathways through which these measures achieve their preventive effect. Extrinsic measures provide immediate protection, but only when worn, as they only provide support for a previously injured ankle and do not target the underlying neuromuscular impairment. Neuromuscular training targets the underlying risk-increasing pathology, but it takes some time before a preventive effect is established. When extrinsic measures are employed during the period of neuromuscular training, the patient benefits

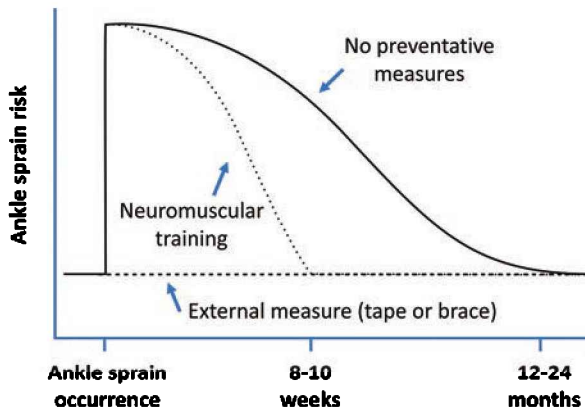


Figure 12.8 Conceptual illustration of optimal ankle re-injury prevention program⁶⁶

from an immediate risk-reducing effect while targeting the underlying causes of an increased recurrence risk.

PREVENTING ACUTE KNEE INJURIES

Preventing major knee injuries such as ACL ruptures should be a priority in many sports, especially those characterised by pivoting movements, sudden changes of direction, and accelerations and decelerations. ACL ruptures are one of the most common types of serious sports injury. For example, a typical elite-level female soccer team can expect one ACL injury every season.⁸⁴ Due to the costly treatment, long rehabilitation time and increased risk of osteoarthritis associated with ACL injuries, they have received a great deal of attention in sports injury prevention research. Much of the research is also relevant to other types of traumatic knee injuries.

Injury mechanisms

Approximately 70% of ACL injuries occur in non-contact situations, often during a cutting manoeuvre or a single-leg landing after a jump.⁸⁵ Cutting or sidestep manoeuvres are associated with dramatic increases in the varus-valgus and internal rotation moments, as well as deceleration.

The typical ACL injury occurs with the knee externally rotated and in 10–30° of flexion when the knee is placed in a valgus position as the athlete takes off from the planted foot and rotates their upper body with the aim of suddenly changing direction. ACL injury mechanisms are discussed in detail in Chapter 35.

Risk factors

The most important intrinsic risk factor for ACL injury is sex, with studies in soccer, basketball, handball and wrestling showing that females have a 2–4 times increased risk compared to their male counterparts.^{86, 87} The reasons

for this are not completely clear. Various researchers have suggested differences in anatomy, and in hormonal and neuromuscular function as potential reasons for the higher injury risk in females. To date, however, there is little evidence linking all these potential intrinsic risk factors to non-contact ACL injuries, and a great deal of controversy exists on the relative importance of the different factors.⁸⁸ Recent studies also suggest that a history of knee ligament injury is a predominant risk factor for a subsequent injury, such as a re-rupture of the ACL graft, an ACL rupture to the contralateral knee, or another severe knee injury.^{33, 89}

An extrinsic risk factor for ACL injury in handball and Australian rules football is high friction between shoes and the playing surface. This can cause the foot to stop abruptly during a cutting or turning manoeuvre, increasing stress on the knee.^{90, 91} The relationship between playing surfaces and injury risk is discussed at the end of this chapter.

Prevention programs

There is strong evidence that acute knee injuries can be reduced by 25–50% using structured exercise programs targeting strength, balance and neuromuscular control.^{92–96} A number of prevention programs are available, all of which are designed to be performed as a structured warm-up prior to training and matches aiming to:

- alter the quality of the movement in the lower limb
- increase core and lower limb stability
- raise awareness of knee position in relation to the foot
- emphasise alignment of the hip, knee and foot in all exercises.

The primary goal of this strategy is to engrain 'safe' movement patterns and discourage those believed to place the knee at high risk of injury (such as hip adduction/knee valgus).

A recent meta-analysis of nine randomised controlled trials investigating injury prevention programs in soccer players demonstrated an overall positive effect in reducing knee injuries in general (26% reduced risk) and a non-significant trend in reducing ACL injuries in particular.⁹⁷ Another meta-analysis, which included randomised and non-randomised trials in a range of sports, found a significant effect of prevention programs in reducing ACL injuries (17% reduced risk). This meta-analysis recommended that for best effect, programs should be aimed at younger athletes, performed for longer than 20 minutes duration, a minimum of twice per week, and include exercise variations and verbal feedback.⁹⁸

Although the International Federation of Association Football (FIFA) 11+ was developed for soccer players, similar multifaceted warm-up programs exist for other sports such as floorball and handball and the same concept can be applied to other sports settings as well.^{93, 100}

11+ A complete warm-up program

with MARIO BIZZINI, ASTRID JUNGHE and JIRI DVOŘÁK

The '11+' injury prevention program was developed by an international group of experts for the International Federation of Association Football (FIFA). It is a good example of a multifaceted 20-minute warm-up program that includes core stability, balance, strength and running exercises (Fig. 12.9). Youth football teams using 11+ as a standard warm-up had

a significantly lower risk of injuries than teams who warmed up as usual. Teams that performed the 11+ at least twice per week had 37% fewer training injuries and 29% fewer match injuries.^{95,99}

The FIFA 11+ has three parts: Part 1 consists of running exercises at a slow speed combined with active stretching,

The 11+

PART 1 RUNNING EXERCISES • 8 MINUTES

- 1 RUNNING STRAIGHT AHEAD**
The exercise is made up of 4 to 10 parts of equal parts, each 5 to 6 m long. The player starts in a ready position, then runs straight ahead for 5 seconds at the speed of the last part of the run. On the way back, you can increase your speed progressively in each part. 2 sets
- 2 RUNNING HIP OUT**
Walk or jog slowly, keeping a straight line of knees to 90° of your knee and rotate your hips outwards. Alternate between left and right legs in a continuous cycle. 2 sets
- 3 RUNNING HIP IN**
Walk or jog slowly, keeping a straight line of knees to 90° of your knee and rotate your hips inwards. Alternate between left and right legs in a continuous cycle. 2 sets
- 4 RUNNING CIRCULAR PARTNER**
Run forwards in a circle in the first part of the run. Shuffle sideways for 90 degrees in the second part. Shuffle backwards for 90 degrees in the third part. Then return back to the circle. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 5 RUNNING SHOULDER CONTACT**
Run forwards in a circle in the first part of the run. Shuffle sideways for 90 degrees in the second part. Shuffle backwards for 90 degrees in the third part. Then return back to the circle. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 6 RUNNING QUICK FORWARDS & BACKWARDS**
At a slow, not quickly, in the second part of the run, you can increase your speed. Run forwards for 10 seconds, then backwards for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets

PART 2 STRENGTH • PLYOMETRICS • BALANCE • 10 MINUTES

LEVEL 1

- 7 THE BENCH STATIC**
Starting position: Lie on your back, supporting yourself on your forearms. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 8 SIDEWAYS BENCH STATIC**
Starting position: Lie on your side with your forearms on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 9 HAMSTRINGS BEGINNER**
Starting position: Lie on your back with your knees bent. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 10 SINGLE-LEG STANCE HOLD THE BALL**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 11 SQUATS WITH TOE RAISE**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 12 JUMPING VERTICAL JUMPS**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets

LEVEL 2

- 7 THE BENCH ALTERNATE LEGS**
Starting position: Lie on your back, supporting yourself on your forearms. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 8 SIDEWAYS BENCH RAISE & LOWER HIP**
Starting position: Lie on your side with your forearms on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 9 HAMSTRINGS INTERMEDIATE**
Starting position: Lie on your back with your knees bent. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 10 SINGLE-LEG STANCE THROWING BALL WITH PARTNER**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 11 SQUATS WALKING LUNGES**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 12 JUMPING LATERAL JUMPS**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets

LEVEL 3

- 7 THE BENCH ONE LEG LIFT AND HOLD**
Starting position: Lie on your back, supporting yourself on your forearms. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 8 SIDEWAYS BENCH WITH LEG LIFT**
Starting position: Lie on your side with your forearms on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 9 HAMSTRINGS ADVANCED**
Starting position: Lie on your back with your knees bent. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 10 SINGLE-LEG STANCE TEST YOUR PARTNER**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 11 SQUATS ONE-LEG SQUATS**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 12 JUMPING BOX JUMPS**
Starting position: Stand on one leg with your feet on the ground. Exercise: Lift your hips and hold them for 10 seconds. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets

PART 3 RUNNING EXERCISES • 2 MINUTES

- 13 RUNNING ACROSS THE PITCH**
Run across the pitch, then back to the start, for 15-20% maximum pace. 2 sets
- 14 RUNNING BOUNDING**
Run with high bounding steps with a high knee lift, landing gently on the ball of your foot. Use an exaggerated swing for each step. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets
- 15 RUNNING PLANT & CUT**
Run 5-6 steps, then plant on the outside leg and cut to change direction. Repeat for each part of the run. Partner to help you keep your feet on the ground and to keep your knees in a straight line. 2 sets

KNEE POSITION CORRECT

KNEE POSITION INCORRECT

MY GAME IS FAIR PLAY

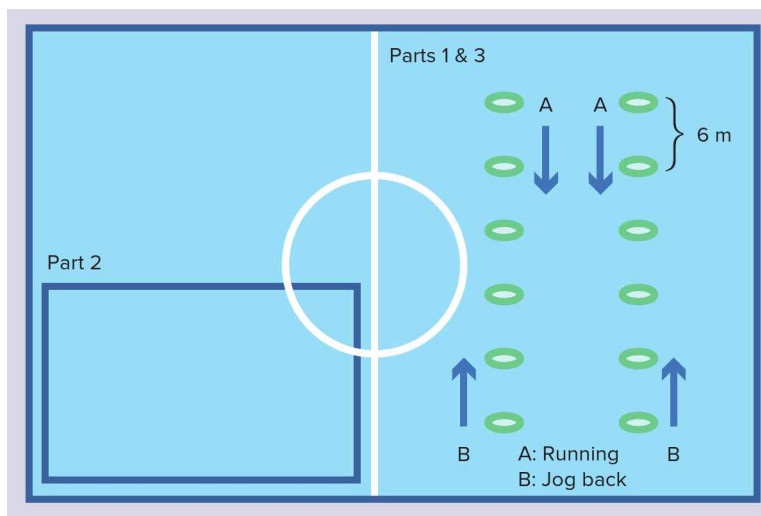
F-MARC FOOTBALL FOR HEALTH

Figure 12.9 The FIFA 11+ poster can be downloaded for free from <http://f-marc.com/fifa-11-plus/>. It is available in nine languages and is provided to clubs free of charge

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and controlled partner contacts; Part 2 consists of six sets of exercises, focusing on core and leg strength, balance and plyometrics/agility, each with three levels of increasing difficulty; and Part 3 consists of running exercises at moderate/high speed combined with planting/cutting movements.

The field set-up course is made up of six pairs of parallel cones, approximately 5–6 m apart. Two players start at the same time from the first pair of cones, jog along the inside of the cones and do the various exercises on the way. After the last cone, they run back along the outside. On the way back, speed can be increased progressively as players warm up.



PART 1: RUNNING EXERCISES

1 RUNNING—STRAIGHT AHEAD

Jog straight to the last cone. Make sure you keep your upper body straight and your hip, knee and foot aligned. Do not let your knee buckle inwards. Run slightly more quickly on the way back. 2 sets.

2 RUNNING—HIP OUT

Jog to the first cone, stop and lift your knee forwards. Rotate your knee to the side and put your foot down. At the next cone repeat the exercise on the other leg. Repeat until you reach the other side of the pitch. 2 sets.

3 RUNNING—HIP IN

Jog to the first cone, stop and lift your knee to the side. Rotate your knee forwards and put your foot down. At the next cone repeat the exercise on the other leg. Repeat until you reach the other side of the pitch. 2 sets.

4 RUNNING—CIRCLING PARTNER

Jog to the first cone. Shuffle sideways towards your partner, shuffle an entire circle around one another (without changing the direction you are looking in), then shuffle back to the first cone. Repeat until you reach the other side of the pitch. 2 sets.

5 RUNNING—JUMPING WITH SHOULDER CONTACT

Jog to the first cone. Shuffle sideways towards your partner. In the middle, jump sideways towards each other to make shoulder-to-shoulder contact. Land on both feet with your hips and knees bent. Shuffle back to the first cone. Repeat until you reach the other side of the pitch. 2 sets.

6 RUNNING—QUICK FORWARD AND BACKWARD SPRINTS

Run quickly to the second cone then run backwards quickly to the first cone, keeping your hips and knees slightly bent. Repeat, running two cones forwards and one cone backwards until you reach the other side of the pitch. 2 sets.

PART 2: STRENGTH, PLYOMETRICS, BALANCE

All exercises have three levels of increasing difficulty.

Players should begin with level 1, which is presented here. Levels 2 and 3 of exercises 7–12 are presented on www.flfa.com/medical and videos are available in the free mobile application GET SET—Train Smarter.

Only when an exercise can be performed without difficulty for the specified duration and number of repetitions should the player progress to the next level.

7.1 THE BENCH—STATIC

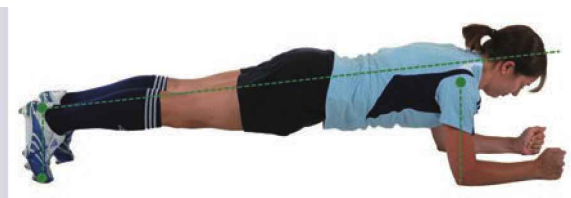
Starting position: Lie on your front, support your upper body with your forearms, elbows directly under your shoulders.

Exercise: Lift up your upper body, pelvis and legs until your

body is in a straight line from head to foot. Pull in stomach and gluteal muscles and hold the position for 20–30 seconds. 3 sets. **Important:** Do not sway or arch your back. Do not move your buttocks upwards.

7.2 ALTERNATE LEGS

7.3 ONE LEG LIFT AND HOLD



continued

8.1 SIDEWAYS BENCH—STATIC

Starting position: Lie on your side with the knee of your lowermost leg bent to 90°. Support yourself on your forearm and lowermost leg, elbow of supporting arm directly under shoulder. **Exercise:** Lift pelvis and uppermost leg until they form a straight line with your shoulder and hold the position for 20–30 seconds. Repeat on other side. 3 sets. **Important:** Keep pelvis stable and do not let it tilt downwards. Do not tilt shoulders, pelvis or leg forwards or backwards.

8.2 RAISE AND LOWER HIP

8.3 WITH LEG LIFT



9.1 HAMSTRINGS—BEGINNER

Starting position: Kneel with knees hip-width apart; partner pins your ankles firmly to the ground with both hands.

Exercise: Slowly lean forwards, while keeping your body straight from the head to the knees. When you can no longer hold the position, gently take your weight on your hands, falling into a press-up position. 3–5 repetitions. **Important:** Do exercise slowly at first, but speed up once you feel more comfortable.

9.2 INTERMEDIATE

9.3 ADVANCED



10.1 SINGLE-LEG STANCE—HOLD THE BALL

Starting position: Stand on one leg, knee and hip slightly bent and hold the ball in both hands. **Exercise:** Hold balance and keep body weight on the ball of your foot. Hold for 30 seconds, repeat on the other leg. Exercise can be made more difficult by lifting the heel from the ground slightly or passing the ball around your waist and/or under your other knee. 2 sets on each leg. **Important:** Do not let your knee buckle inwards. Keep your pelvis horizontal and do not let it tilt to the side.

10.2 THROWING BALL WITH PARTNER

10.3 TEST YOUR PARTNER



11.1 SQUATS WITH TOE RAISE

Starting position: Stand with feet hip-width apart, hands on your hips. **Exercise:** Slowly bend hips, knees and ankles until your knees are flexed to 90°. Lean upper body forwards. Straighten upper body, hips and knees and stand up on your toes. Slowly lower down again, and straighten up slightly more quickly. Repeat for 30 seconds. 2 sets. **Important:** Do not let your knee buckle inwards. Lean upper body forwards with a straight back.

11.2 WALKING LUNGES

11.3 LEG SQUATS



12.1 JUMPING—VERTICAL JUMPS

Starting position: Stand with your feet hip-width apart, hands on your hips. **Exercise:** Slowly bend hips, knees and ankles until your knees are flexed to 90°. Lean upper body forwards. Hold this position for 1 second, then jump as high as you can, and straighten whole body. Land softly on the balls of your feet. Repeat for 30 seconds. 2 sets. **Important:** Jump off both feet. Land gently on the balls of both feet with your knees bent.

12.2 LATERAL JUMPS

12.3 BOX JUMPS

PART 3: RUNNING EXERCISES

13 RUNNING—ACROSS THE PITCH

Run approximately 40 m across the pitch at 75–80% of maximum pace, then jog the rest of the way. Keep your upper body straight. Your hip, knee and foot are aligned. Do not let your knees buckle inwards. Jog easily back. 2 sets.

14 RUNNING—BOUNDING

Take a few warm-up steps then take 6–8 high bounding steps with a high knee lift and then jog the rest of the way. Lift the knee of the leading leg as high as possible and swing the opposite arm across the body. Keep your upper body straight. Land on the ball of the foot with the knee bent and spring. Do not let your knee buckle inwards. Jog back easily to recover. 2 sets.

15 RUNNING—PLANT AND CUT

Jog 4–5 steps straight ahead. Then plant on the right leg and cut to change direction to the left and accelerate again. Sprint 5–7 steps (80–90% of maximum pace) before you



decelerate and plant on the left foot and cut to change direction to the right. Do not let your knee buckle inwards. Repeat the exercise until you reach the other side of the pitch, then jog back. 2 sets.

PRACTICE PEARL

The key point of the program is to use the proper technique during all exercises. Pay close attention to correct posture and body control, including straight-leg alignment, knee-over-toe position and soft landings. The FIFA 11+ should be completed as a complete warm-up at least 2–3 times per week; it should take approximately 20 minutes to complete.

See the instructional video of each exercise on <http://f-marc.com/fifa-11-plus/>. The FIFA 11+ poster can be downloaded for free. It is available in nine languages and is provided to clubs free of charge.

PREVENTING OVERUSE INJURIES

Overuse injuries are the predominant injury type in sports involving long, monotonous training sessions such as cycling, swimming and running, as well as in technical sports that involve the repetition of similar movement patterns such as throwing, jumping and kicking.¹¹ They occur among athletes of all levels, particularly in elite athletes and young, dedicated athletes whose total training and competition load increases rapidly.¹⁰¹

Although a majority of sports injury prevention research has focused on acute injuries, evidence for overuse injury prevention is increasing.¹⁰² Similar to acute injuries, prevention strategies for overuse injuries can target modifiable intrinsic risk factors such as

flexibility, strength and neuromuscular control, as well as extrinsic risk factors such as shoes or sports equipment. As the 'inciting event' for overuse injuries is cumulative tissue overload that occurs over time, close monitoring of athletes' training and competition loads may be a particularly effective prevention strategy. There is also an opportunity for secondary prevention of overuse injuries via early identification and rapid initiation of appropriate treatment.

Stretching

Although stretching has long been promoted as an injury prevention method, recent systematic reviews conclude that there is no evidence to support its efficacy.^{103–105} However, targeted stretching of glenohumeral joint

internal rotation remains a popular prevention strategy in throwing athletes, and formed one part of a multimodal training program shown to reduce the rate of overuse shoulder injuries in elite handball players.¹⁰⁶

Structured training programs

As discussed previously in this chapter, there is strong evidence that structured multimodal training programs targeting muscle strength and neuromuscular control (such as the FIFA 11+) are effective in preventing acute lower limb injuries in team sports. This approach has also been shown to be effective in preventing lower limb overuse injuries,^{93, 95, 107} particularly among adolescent athletes whose risk of overuse injury is almost halved by completing injury prevention programs.¹⁰³

PRACTICE PEARL

Structured warm-up programs consisting of strength and neuromuscular control exercises have been shown to halve the rate of lower limb overuse injuries among adolescent athletes.

These promising results have not been consistently replicated in adult populations. A number of studies have failed to show an effect of general or targeted prevention training on lower limb overuse injuries among amateur athletes, professional athletes and military recruits.^{78, 108-112} For example, a study of 209 professional soccer players found that eccentric training programs commonly used to treat Achilles and patellar tendinopathy were ineffective when used as an across-the-board prevention strategy.¹¹² However, one randomised controlled trial found that patellofemoral pain could be prevented among military recruits using a daily exercise program consisting of gluteal and quadriceps strengthening exercises and lower limb stretching exercises.¹¹³ Similarly, a recent randomised trial found that a 10-minute warm-up program consisting of glenohumeral internal rotation stretching, rotator cuff strengthening and scapular stability exercises reduced the prevalence of shoulder injuries among elite-level handball players.¹⁰⁶ Research in this field is ongoing.

Technique modification

Ensuring optimal sporting technique is likely to be an effective way of preventing overuse injuries, particularly in sports involving repetitive loading patterns. For example, runners should avoid excessive contralateral hip drop to prevent lower limb injuries and lower back pain (Chapter 8), and tennis players should aim for double-handed backhand shots to prevent 'tennis elbow'

(Chapter 9). Further examples of sport-specific technique errors linked to injury are covered in Chapter 9.

Nutritional strategies to prevent stress fractures

Inadequate levels of calcium and vitamin D, which play an essential role in bone mineralisation, homeostasis and remodelling, are thought to be risk factors for stress fractures (Chapter 4).¹¹⁴ This may be a particular problem in individuals with inadequate dietary intake or with limited sunlight exposure (due to geographical or cultural reasons), dark skin, or medical conditions that interfere with fat absorption (such as Crohn's disease).¹¹⁵

A randomised controlled trial involving female military recruits found that daily supplementation with 100 mg calcium and 800 IU vitamin D reduced the incidence of stress fractures by 20%.¹¹⁶ This finding is supported by cohort studies of athletes which have shown that high dietary intake of calcium and of vitamin D are associated with lower stress fracture risk.^{117, 118} For example, one study of female endurance runners found that for every cup of skim milk consumed per day, the risk of stress fractures was reduced by 63%.¹¹⁷

According to current recommendations, 1000 to 1200 mg of calcium per day is required for optimising bone health. Although there is debate regarding optimal vitamin D status, serum 25-hydroxy vitamin D levels should be at least 50 nmol/mL and should ideally be between 90 and 100 nmol/mL.¹¹⁵ The current recommended daily intake of vitamin D is 600 to 800 IU; however, individual needs may vary according to many factors including sun exposure, and some suggest that up to 2000 IU is indicated in high-risk groups.¹¹⁵

It should be noted that bisphosphonates (osteoclast inhibitors), which are commonly used in the treatment of osteoporosis and which may be useful in the treatment of stress fractures,^{114, 119} are not effective in preventing bone stress injuries.¹²⁰

Modifying extrinsic risk factors

In certain sports, injuries are commonly attributed to extrinsic factors such as shoes, insoles and sporting equipment. The extent to which these factors play a role in injury prevention remains a source of much debate.

Shoe selection

Individual prescription of different shoe types (such as cushioning, motion control and stability shoes) based on static foot alignment is commonly promoted as an injury prevention method, particularly among shoe retailers.¹²¹ However, a recent meta-analysis of randomised controlled trials suggests that this approach does little to prevent

injuries.¹²² In several large trials conducted on military recruits, individualised prescription of shoe type was compared to all subjects receiving a stability shoe.¹²³⁻¹²⁵ No studies were able to demonstrate a difference in the incidence of overuse injuries between groups.

Barefoot/minimalist running

It has been proposed that running barefoot or in 'minimalist' shoes designed to replicate barefoot running may prevent lower limb overuse injuries.^{126, 127} This is because several of the common biomechanical differences seen in barefoot running, such as reduced peak ground reaction forces and increased ankle plantarflexion and knee flexion at foot strike (Chapter 8),¹²⁸ are theoretically protective against some injury types. However, this may be an over-simplification of a complex relationship. Although the loads on some structures such as the knee and hip joints may be reduced in barefoot running, other structures such as the forefoot, ankle, Achilles tendon and calf muscles experience increased loading and may be at greater risk of injury. Also, certain individuals fail to adapt their running technique and experience *increased* vertical ground reaction forces when barefoot running.

There is currently no evidence that barefoot or minimalist-shoe running is an effective prevention intervention.¹²⁹ In fact, one study which randomly assigned normal (neutral), partial-minimalist and full-minimalist shoes to 103 runners found that the groups using minimalist shoes had injury rates up to three times higher than those in normal neutral shoes.¹³⁰ The use of barefoot or minimalist shoes is therefore likely to have more value as a targeted strategy in the management of certain types of running injuries (as discussed in Chapter 8) than as a generalised injury prevention intervention.

Orthotic insoles

The preventive effects of prefabricated and custom-made orthotic insoles have been investigated in 10 different studies of military personnel, including a total of 4788 subjects.¹⁰⁵ Seven studies found that insoles had a significant preventive effect for a range of lower limb overuse conditions, such as stress fractures, medial tibial stress syndrome, patellofemoral pain and Achilles tendinopathy. Based on these findings, several high-quality meta-analyses have supported the use of orthotic insoles for injury prevention.^{104, 105, 131} However, the effects of orthotic insoles in athletes remain unknown, and the exact mechanism by which they prevent injury remains contentious.¹³¹

Sports equipment

Careful selection of sports equipment may play an important role in overuse injury prevention. Specific

examples of equipment-related factors related to injury are discussed in Chapter 9.

MANAGING LOAD TO PREVENT INJURY

with TIM GABBETT

The concept of load management was introduced in Chapter 10 with a focus on athletic performance enhancement. Here, we revisit load management with greater focus on its increasingly recognised role in injury prevention. Of course, this is an artificial separation. In reality, injury prevention and performance enhancement are intimately linked.

The relationship between load and injury

High training and competition loads have been linked to increased injury risk in a wide variety of sports, including Australian rules football, baseball, basketball, cricket, rugby league, rugby union, rowing, soccer, swimming, volleyball and water polo.¹³² This may not only be due to an increased rate of overuse injuries, but also of acute injuries, as overloaded athletes can have impaired neuromuscular control, reaction time and decision-making ability.

In some studies, high absolute loads have been shown to increase injury risk, which has led to certain sport-specific recommendations or regulations on the total amount of training and competition that athletes should perform. For example, in little league baseball there are specific age-based rules on the maximum number of pitches that players can perform per day and the minimum amount of rest required between matches.¹³³ The effects of these regulations remain undocumented. However, they have been well accepted by the sport and are often credited as having reduced the rate of injury among young baseball pitchers. Similar recommendations have been published by the English Cricket Board in an effort to prevent injuries in fast bowlers.

Although limiting the total amount of training and competition is likely to reduce injuries in many sports, that approach is unlikely to be accepted by elite-level athletes and coaches, whose primary concern is to train hard to maximise performance. The approach also fails to account for the large individual variation in load tolerance between athletes. Any given team will contain 'fragile' athletes as well as 'robust' ones who can tolerate very high loads. In fact, some studies have shown that high training loads may indeed be protective against injury, most probably due to enhanced physical resilience. This is the 'training-injury prevention paradox'.¹³⁴

Rather than limiting the absolute amount of training and competition that athletes can perform, recent attention has focused on developing ways by which

athletes can safely achieve and maintain high loads while minimising injury risk. These include:

1. managing the rate of load increase
2. managing the acute:chronic load ratio, and
3. adjusting training based on athletes' responses to external loads.

In order to monitor the loads athletes are subjected to, they must be quantified. The term 'external load' is often used to refer to loads placed upon the athlete, such as their training volume and intensity, the number of repetitions of certain movements (e.g. throws or jumps) or the number of impacts they are subjected to. Recent technological advances such as GPS, power monitors and wearable microsensors have greatly improved the ability to quantify external loads.¹³⁵

The term 'internal load' is used to refer to the athlete's response to an external training or competition load. It can be measured using subjective measures such as the rate of perceived exertion (RPE) or standardised wellness questionnaires,^{136, 137} or using objective measures such as heart rate, physical performance tests, blood markers and heart rate variability.¹³⁷

Both external and internal loads, or indeed a combination of the two (e.g. training duration $\times$ RPE), can be used to monitor the rate of load increase and the acute:chronic load ratio.

Monitoring the rate of load increase

A number of studies have shown that large week-to-week differences in training load increase injury risk, irrespective of the total training load.¹³⁸⁻¹⁴⁰ For example, a study of Australian Football League players by Piggott et al. showed that 40% of injuries were associated with a change in weekly training load of greater than 10%, compared to the preceding week.¹³⁸ Similarly, Gabbett found that when rugby league players' weekly training loads increased by less than 10%, their risk of injury was under 10%. However, when training load was increased by 15% or more, players' injury risk escalated to between 21% and 49% (Fig. 12.10).¹³⁴

Of course, the rate at which week-to-week changes in load can be tolerated will vary between individual athletes and across sports. For example, Nielsen et al. found that novice runners could tolerate a weekly progression in training volume of between 20% and 25% in the first 10 weeks of a running program, without a dramatic increase in injury risk.¹⁴⁰ Nevertheless, based on the current evidence, limiting week-to-week training load increases to 10% seems to be a good rule of thumb.

Monitoring the acute:chronic load ratio

The acute:chronic load ratio reflects an athlete's training status at a given point in time, taking into account

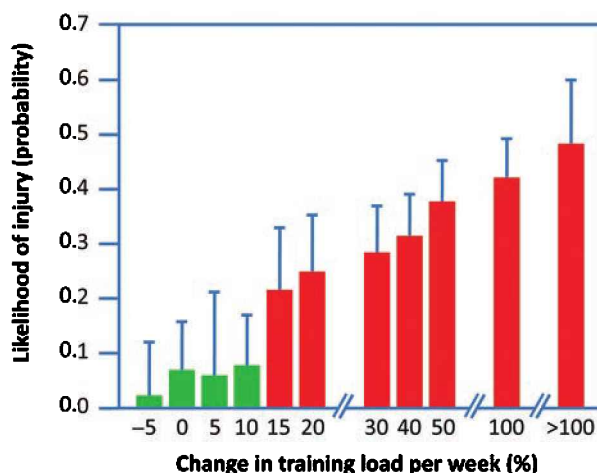


Figure 12.10 Likelihood of injury among professional rugby league players with different weekly changes in training load. Note the dramatic increase in risk when week-to-week loading increases by $\geq 15\%$. FIGURE ADAPTED FROM GABBETT¹³⁴

their training completed over the previous week (their acute load, synonymous with 'fatigue') and their training completed over the previous 4 weeks (their chronic load, synonymous with 'fitness').

PRACTICE PEARL

The acute:chronic load ratio has been linked to both performance and injury risk. An athlete with a high chronic and low acute load will be fit, well rested and ready to perform. Conversely, an athlete with a low chronic and high acute load will be fatigued and have an increased risk of injury.

Studies in rugby league,^{141, 142} cricket,¹⁴³ Australian rules football,¹⁴⁴ Gaelic football¹⁴⁵ and soccer¹⁴⁶ have demonstrated that athletes with high acute:chronic load ratios have an increased risk of injury. Modelling data from several of these studies, Gabbett showed that ratios exceeding 1.5 represented a 'danger zone' where injury risk was substantially increased.¹³⁴ In contrast, ratios between 0.8 and 1.3 represented a 'sweet spot' where injury risk was minimised (Fig. 12.11).

Although the exact thresholds may vary in other sports, the underlying concept is likely to remain the same. In order to minimise their injury risk, athletes should build up training loads gradually, closely monitoring their acute:chronic load ratio. This is especially relevant after a period of reduced training (e.g. due to illness or injury) or when resuming training after an off-season break.¹⁴⁷

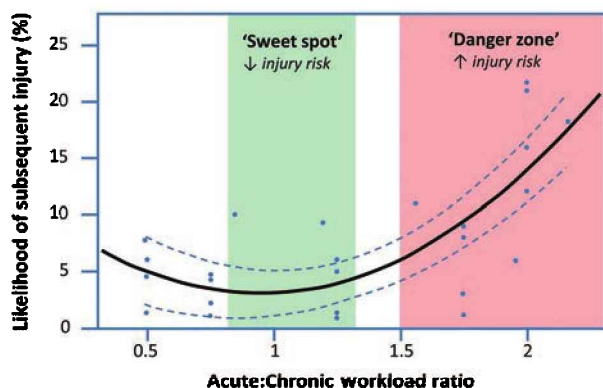


Figure 12.11 The relationship between the acute:chronic ratio and injury, modelled of using data from cricket, Australian rules football and rugby league
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The acute:chronic load ratio helps us understand how two athletes performing the same training session may have completely different injury risk, depending on the amount and consistency of training they have performed over the past month. Similarly, it provides athletes and coaches with a guide to what is an acceptable level of training progression to minimise injury risk.

Monitoring athletes' response to load—the traffic-light approach

Continuous monitoring of athletes' response to training is common practice in many high-level team sports, and may be useful in preventing overtraining, illness and injury. In this context, the goal of daily monitoring is to identify early signs of deleterious overtraining such that appropriate adjustments can be made to the athletes'

Calculating the acute:chronic load ratio

The acute:chronic load ratio can be calculated using a wide range of load measures, such as GPS and accelerometer-derived metrics,¹⁴⁶ throw counts,¹⁴³ and session RPE (training session duration multiplied by the athlete's rate of perceived exertion; see Chapter 10).

Table 12.3 shows an athlete's training data over 12 weeks. In this example, the acute load represents the

cumulative session-RPE data for all training sessions completed by the athlete each week. The chronic load represents the rolling average of the previous 4 weeks' acute loads (i.e. in week 4, the chronic load is the average of weeks 1, 2, 3 and 4. In week 5, it is the average of weeks 2, 3, 4 and 5, and so on). The acute:chronic load ratio is calculated each week by dividing the acute load by the chronic load.

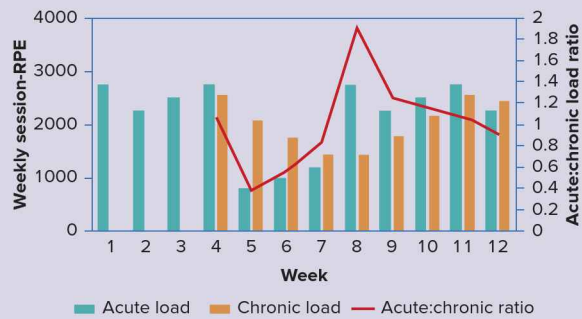
Table 12.3 An example of calculating the acute:chronic load ratio based on an athlete's session-RPE data over 12 weeks. Note that 4 weeks of data are necessary to calculate the chronic loads and acute:chronic load ratio

Week	Acute load (current week)	Chronic load (average of previous 4 weeks)	Acute:chronic load ratio
1	2750		
2	2250		
3	2500		
4	2750	2562.5	1.07
5	800	2075	0.39
6	1000	1762.5	0.57
7	1200	1437.5	0.83
8	2750	1437.5	1.91
9	2250	1800	1.25
10	2500	2175	1.15
11	2750	2562.5	1.07
12	2250	2437.5	0.92

continued

In this example, the athlete had a period of reduced training in weeks 5, 6 and 7—perhaps due to a minor injury. The rapid resumption of normal training in week 8 leads to a spike in the acute:chronic load ratio into the ‘danger zone’ above 1.5 (Fig. 12.12), indicating an increased risk of subsequent injury.

Figure 12.12 Monitoring an athlete’s acute:chronic load ratio, based on data presented in Table 12.3



training program. We refer to this as the ‘traffic light approach’ as, based on their daily monitoring results, athletes are often categorised as: able to complete full training (green light); needing reduced training (orange light); or needing complete rest (red light).

Measures that have been used in efforts to detect overtraining can be broadly classified into objective and subjective measures (Table 12.4).

It is not easy for clinicians to decide on the best method of athlete monitoring. Each approach has its own strengths and limitations, and despite an increasing number of commercial claims to the contrary, there is no magic formula to predict which athletes will get sick or injured.

A recent systematic review found that subjective and objective measures generally did not correlate, with subjective measures being more responsive to changes in acute and chronic training loads.¹³⁶ As shown in Table 12.4, a number of validated questionnaires can be used to monitor athletes for signs of overtraining. However, these consist of 22–76 questions, making them too time consuming for daily administration.¹⁵⁸ Therefore, teams often use short questionnaires consisting of 5–7 Likert-scale questions for

daily monitoring, with more comprehensive questionnaires administered less frequently.¹⁵⁹ The content of these short questionnaires varies; however, the following subscales have recently been identified as being responsive to acute changes in load:¹³⁶

- non-training stress
- fatigue
- physical recovery
- general health/wellbeing
- being in shape.

Irrespective of the monitoring measures used, the main challenges facing the clinician include:

- establishing each athlete’s baseline: their normal training response needs to be known before abnormal responses can be identified
- differentiating between intended and deleterious overtraining: coaches will often schedule hard training sessions to stimulate a training response (Chapter 10).

Due to these challenges, monitoring for overtraining remains as much art as science, and necessitates a close communication with athletes and coaching staff.

Table 12.4 Examples of measures used to monitor athletes for signs of deleterious overtraining

Objective measures
Heart rate (HR) measures (e.g. HR versus RPE, ¹⁴⁸ HR recovery ¹⁴⁹ and HR variability ¹⁵⁰)
Biochemical, hormonal and immunological markers (from blood or saliva) ^{137, 151}
Physical performance tests (e.g. vertical jump height ¹⁵²)
Psychomotor speed ¹⁵³
Subjective measures
Profile of mood states (POMS) ^{154, 155}
Recovery stress questionnaire for athletes (RESTQ-S) ¹⁵⁶
Daily analyses of life demands of athletes (DALDA) ¹⁵⁷

PROTECTIVE EQUIPMENT

Protective equipment has been designed to shield various parts of the body against injury without interfering with sporting activity. Protective equipment can also be used on return to activity after injury in situations where direct contact may aggravate the injury.

Helmets are mandatory in certain sports such as motor racing, motor cycling, cycling, ice hockey, horse-riding and American football. In other sports, the use of helmets is not universally accepted, such as in rugby and skateboarding.

Other protective equipment commonly worn includes: mouth guards in most collision sports; shoulder pads in American football and rugby; chest, forearm and groin protectors in ice hockey; knee pads when playing on artificial surfaces or while rollerblading; wrist guards in rollerblading and snowboarding; and shin pads in soccer and hockey. It is important that protective equipment fits correctly.

The use of protective equipment may increase an athlete's confidence. This may help prevent injuries, but may also lead the athlete to take bigger risks or play more aggressively.

APPROPRIATE SURFACES

with JOHN ORCHARD

Playing surfaces remain under the injury prevention spotlight, as differing traction and shock absorption properties have long been linked to injury risk. The association between playing surface and injury risk was first proposed in handball as early as 1990.¹⁶⁰ Later, a

large epidemiological study compared the ACL injury rate between two different floor types—wooden floors (parquet, generally having lower friction) and artificial floors (generally having higher friction).¹⁶¹ The results indicated that the risk of ACL injury among female handball players is higher on high-friction artificial floors than on wooden floors. However, other factors also play a significant role for shoe surface friction, principally shoe type and floor maintenance.

In Australian rules football, Orchard and colleagues noted the greater rate of ACL injuries in the northern (warmer) climates.¹⁶² Although it was tempting to attribute this to drier weather, and thus ground hardness, that hypothesis was not supported by data from American football teams where games were played on natural grass.¹⁶³ Further analysis of both the Australian and US data suggested that the type of grass, and thus, the tightness of the thatch, may influence ACL risk; the more northern Australian rules football venues had grass types that permitted excessive shoe-surface traction (Fig. 12.13a-d).

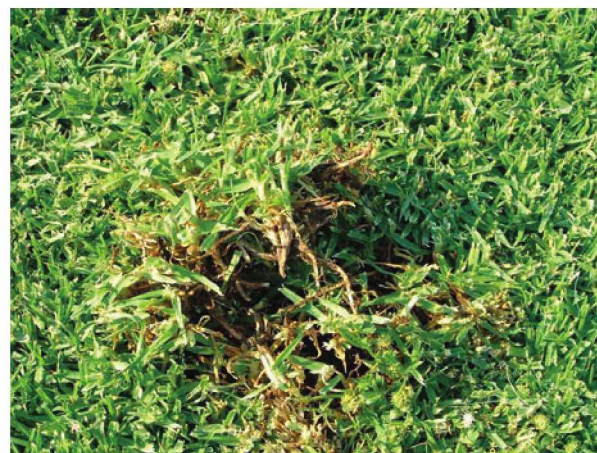
According to turf authority McNitt, perennial rye grass is associated with lower shoe-surface traction than Kentucky blue grass or Bermuda grass because it creates less thatch.¹⁶⁴ These studies suggest that rye grass generally offers a safer surface with respect to ACL injuries for football than some other grasses.

Natural grass versus artificial turf

The link between artificial turf and injury risk has long been controversial. Early generations of turf, introduced in the 1960s, were linked to an increased risk due to their



(a)



(b)

Figure 12.13 Four different types of grasses that provide the surface for Australian rules football and have been associated with different rates of ACL injury (a) Bermuda ('couch') grass surface, showing a thick thatch layer between grass leaves and soil (b) Kikuyu grass, also showing a thick thatch layer



(c)



(d)

Figure 12.13 (cont.) (c) Rye grass surface, showing a minimal thatch layer. This is probably a safer surface than the others, as the blades or cleats of the football boot are less likely to be 'gripped' by the surface (d) Annual blue grass surface, showing a moderate thatch layer

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having minimal padding and high friction coefficients.¹⁶⁵ However, new turf types are continuously being developed, with newer types (third and fourth generation) having properties closer to natural grass.¹⁶⁶ Research into the injury risk of third- and fourth-generation turf

is conflicting, and in some cases may have been biased by industry funding. Nevertheless, recent systematic reviews have concluded that there is little evidence of any difference in overall injury risk between natural and artificial turf.^{165, 167}

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e